

Attention mechanisms

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CAS Building ML/AI Applications, ETH Zurich, weekend 2

Friday 18 September 2026

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Section 1

Attention

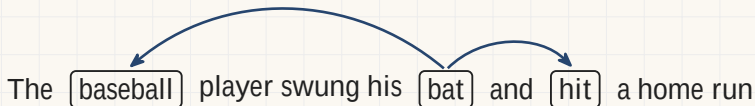
Attention

Words can find their meaning in a sentence by **attending to** other words!

The baseball player swung his bat and hit a home run

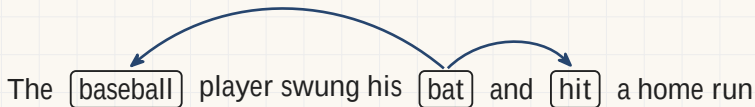
Attention

Words can find their meaning in a sentence by **attending to** other words!



Attention

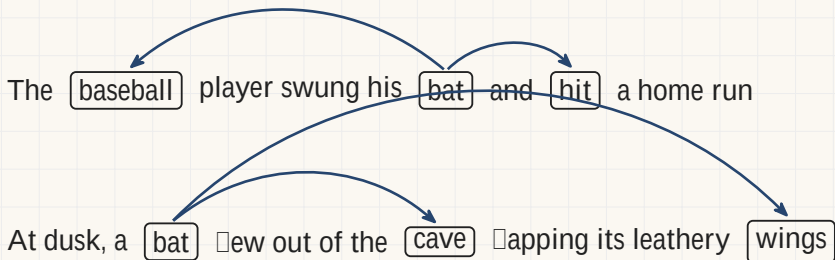
Words can find their meaning in a sentence by **attending to** other words!



At dusk, a bat flew out of the cave flapping its leathery wings

Attention

Words can find their meaning in a sentence by **attending to** other words!



Attention as information retrieval

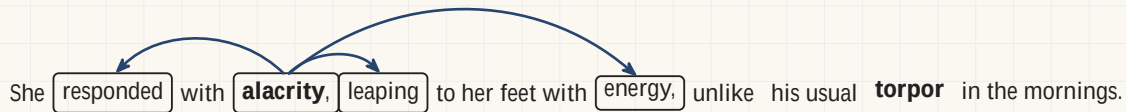
- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?

Attention as information retrieval



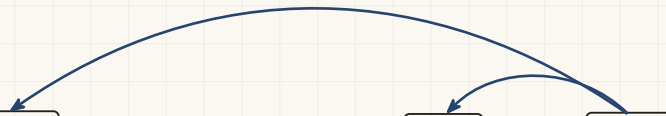
- She responded with **alacrity**, leaping to her feet with energy, unlike his usual **torpor** in the mornings.
- What do the words “alacrity” and “torpor” mean?
- Alacrity: brisk and cheerful readiness.
- Torpor: a state of physical or mental inactivity; lethargy.

Attention as information retrieval



Attention as information retrieval

She responded with **alacrity**, leaping to her feet with energy, **unlike** his usual **torpor** in the mornings.



Defining attention mechanisms

- We **define** a **value** representation for each word.

Defining attention mechanisms

- We **define** a **value** representation for each word.
- We let other words **enrich** their representations by **attending** to other words.

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.



streept kat



bootet kat

Creating embeddings for words

- Recall the Wuggish game. Every sentence describes at most three things.



streept kat



vennomus snaak wudden chaar



bootet kat



streept snaak streept kat wudden chaar

Creating representations

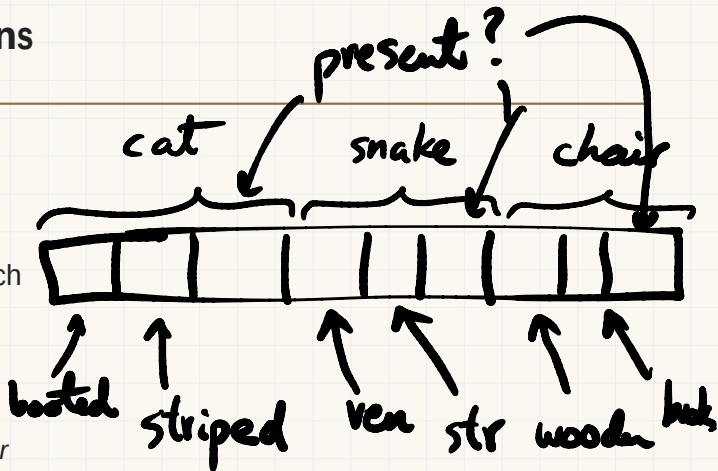
- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

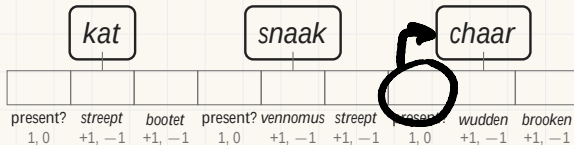

vennomus streept snaak wudden chaar


kat brooken chaar



Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?




vennomus streept snaak wudden chaar


kat brooken chaar

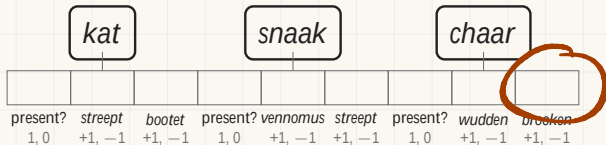
wudden
 1 0 1 0 1 0 1 0 1 0 1 0

brooken
 0 0 1 0 0 1 0 1 0 1 1

wudden & brooken
 1 1 1 1 1 1 1 1 1 1

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?




vennomus streept snaak wudden chaar

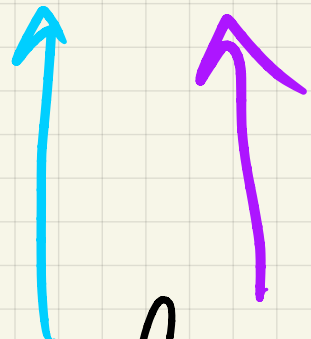

kat brooken chaar

char

venomous

striped

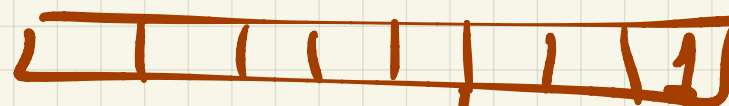
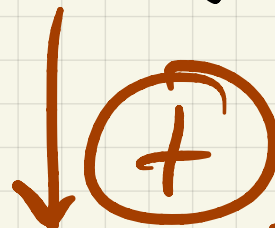
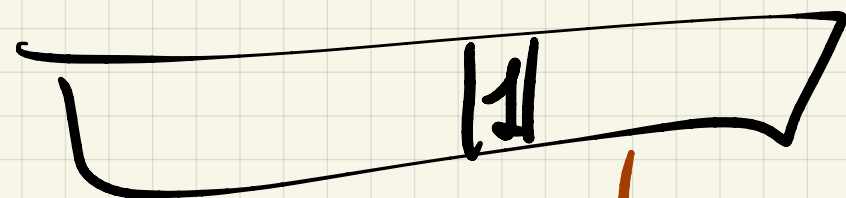
broken snake



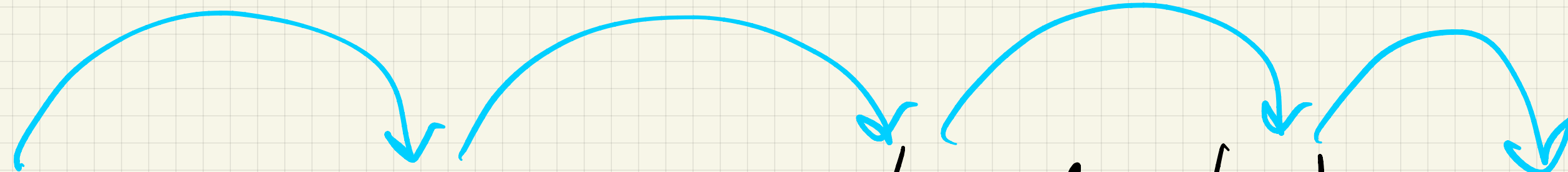
char

What is my meaning?

"I'm present"



X



Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

	kat			snaak			chaar		
	present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1
A	0	0	0	1	+1	+1	1	+1	-1



vennomus streept snaak wudden chaar



kat brooken chaar

Creating representations

- There are 8 words.
- There are 124 possible sentences.
- How can we characterize each sentence?

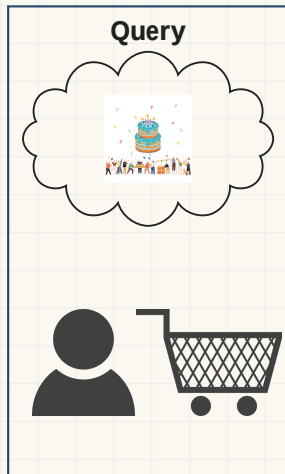
kat			snaak			chaar		
present?	<i>streept</i>	<i>bootet</i>	present?	<i>vennomus</i>	<i>streept</i>	present?	<i>wudden</i>	<i>brooken</i>
1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1	1, 0	+1, -1	+1, -1

A	0	0	0	1	+1	+1	1	+1	-1
B	1	-1	-1	0	0	0	1	-1	+1


vennomus streept snaak wudden chaar


kat brooken chaar

Attention as information retrieval



Attention as information retrieval



Attention as information retrieval



2

Section 2

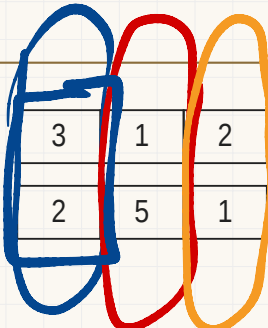
The inner product

The inner product

a

3	1	2
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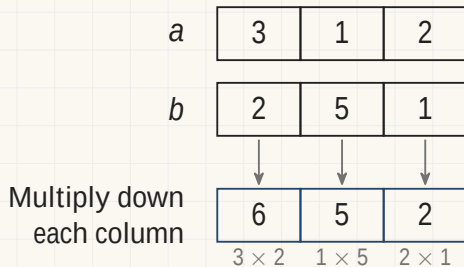
The inner product



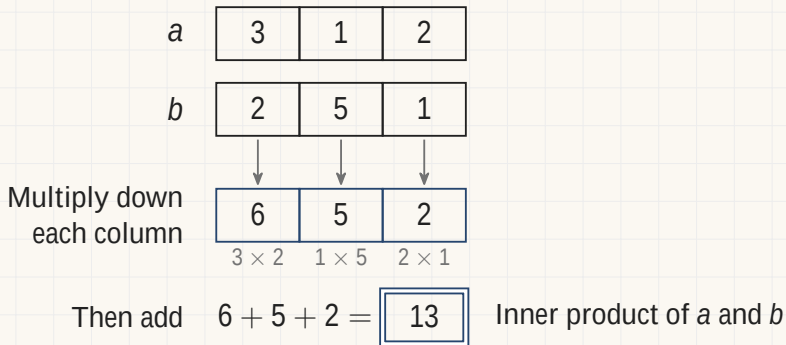
a	3	1	2
b	2	5	1

$$3 \times 2 + 1 \times 5 + 2 \times 1 =$$
$$6 + 5 + 2 = 13 \quad \text{😊}$$

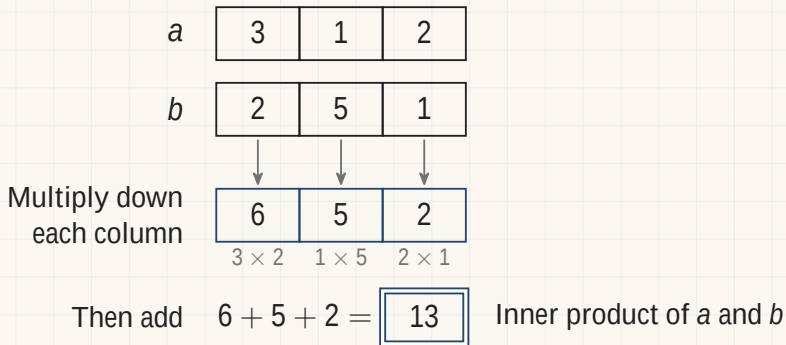
The inner product



The inner product



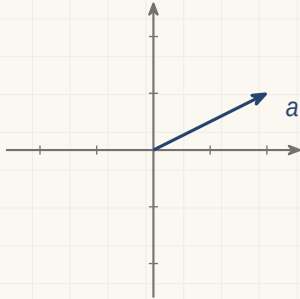
The inner product



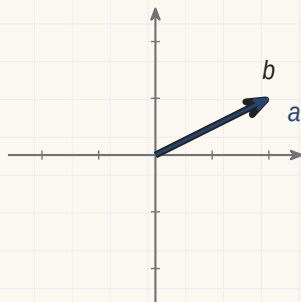
$$\text{Inner product of } a \text{ and } b = a_1b_1 + a_2b_2 + a_3b_3$$

The inner product measures similarity

$$\underline{a = (2, 1)}$$



The inner product measures similarity



$$\underline{a = (2, 1)}$$

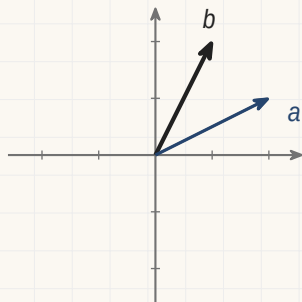
$$\underline{b = (2, 1)}$$

$$2 \times 2 + 1 \times 1 = 5$$



Same direction

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

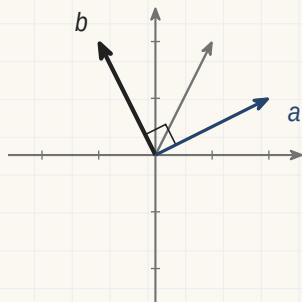
Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

Close directions

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1)$$

$$2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2)$$

$$2 \times 1 + 1 \times 2 = 4$$

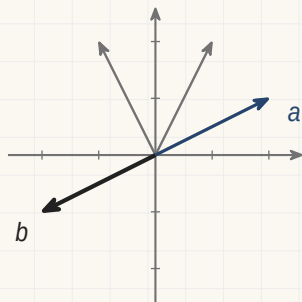
Close directions

$$b = (-1, 2)$$

$$2 \times (-1) + 1 \times 2 = 0$$

At right angles

The inner product measures similarity



$$a = (2, 1)$$

$$b = (2, 1) \quad 2 \times 2 + 1 \times 1 = 5$$

Same direction

$$b = (1, 2) \quad 2 \times 1 + 1 \times 2 = 4$$

Close directions

$$b = (-1, 2) \quad 2 \times (-1) + 1 \times 2 = 0$$

At right angles

$$b = (-2, -1) \quad 2 \times (-2) + 1 \times (-1) = -5$$

Opposite directions

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
Query of <i>kat</i>	1	0	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product
Query of <i>kat</i>	1	0	0	
Key of <i>kat</i>	1	0	0	1
Key of <i>bootet</i>	1	0	0	1
Key of <i>wudden</i>	0	0	1	0
Key of <i>streept</i>	1	1	0	1

The inner product of zeros and ones

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	Inner product	
Query of <i>kat</i>	1	0	0		
Key of <i>kat</i>	1	0	0	1	It counts the slots where both have a 1
Key of <i>bootet</i>	1	0	0	1	
Key of <i>wudden</i>	0	0	1	0	
Key of <i>streept</i>	1	1	0	1	

3

Section 3

The attention mechanism

An attention map

wudden

kat

vennomus

chaar

bootet

snaak

brooken

An attention map

Queries

	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0
<i>kat</i>	1	0	0
<i>vennomus</i>	0	0	0
<i>chaar</i>	0	0	1
<i>bootet</i>	0	0	0
<i>snaak</i>	0	1	0
<i>brooken</i>	0	0	0

An attention map

	Queries			Keys		
	<i>kat</i>	<i>snaak</i>	<i>chaar</i>	<i>kat</i>	<i>snaak</i>	<i>chaar</i>
<i>wudden</i>	0	0	0	0	0	1
<i>kat</i>	1	0	0	1	0	0
<i>vennomus</i>	0	0	0	0	1	0
<i>chaar</i>	0	0	1	0	0	1
<i>bootet</i>	0	0	0	1	0	0
<i>snaak</i>	0	1	0	0	1	0
<i>brooken</i>	0	0	0	0	0	1

An attention map

I'm this type of object
HELP!

Queries

Keys

Values

kat
snaak
chaar

kat
snaak
chaar

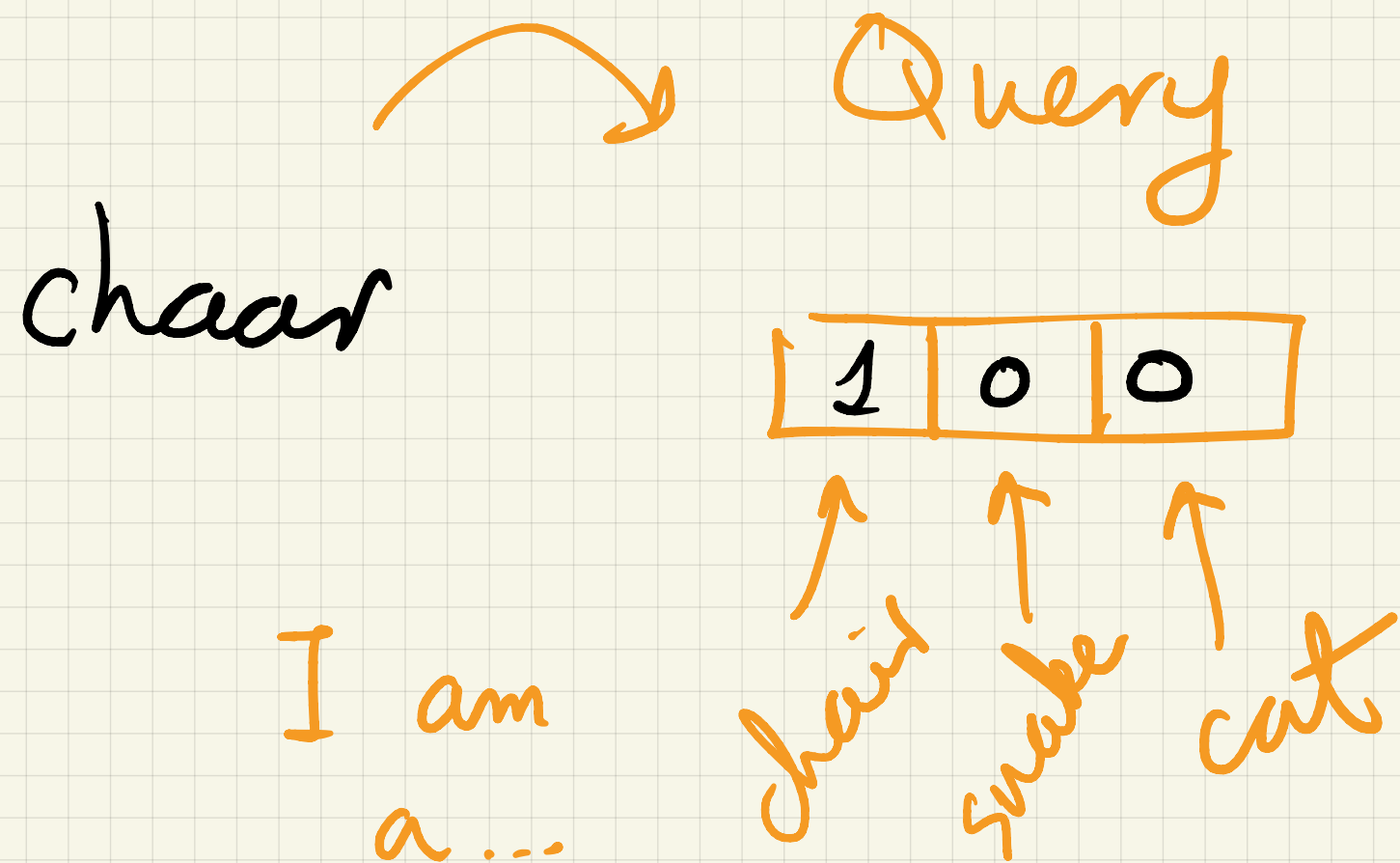
kat snaak chaar
present? present? present?
streept vennomus wudden
bootet streept brooken

I can provide
meaning to
this type
of objs

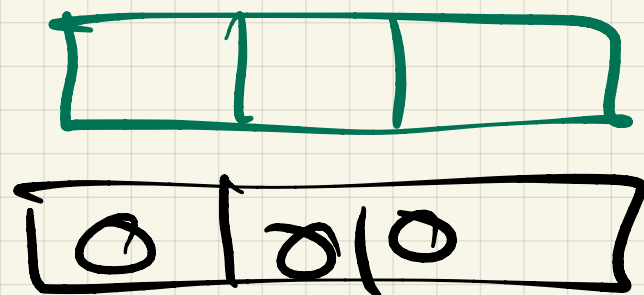
wudden	0	0	0	0	0	1	0	0	0	0	0	0	0	+2	0
kat	1	0	0	1	0	0	1	-1	-1	0	0	0	0	0	0
vennomus	0	0	0	0	1	0	0	0	0	0	+2	0	0	0	0
chaar	0	0	1	0	0	1	0	0	0	0	0	0	1	-1	-1
bootet	0	0	0	1	0	0	0	0	+2	0	0	0	0	0	0
snaak	0	1	0	0	1	0	0	0	0	1	-1	-1	0	0	0
brooken	0	0	0	0	0	1	0	0	0	0	0	0	0	0	+2

Ingredients for attention

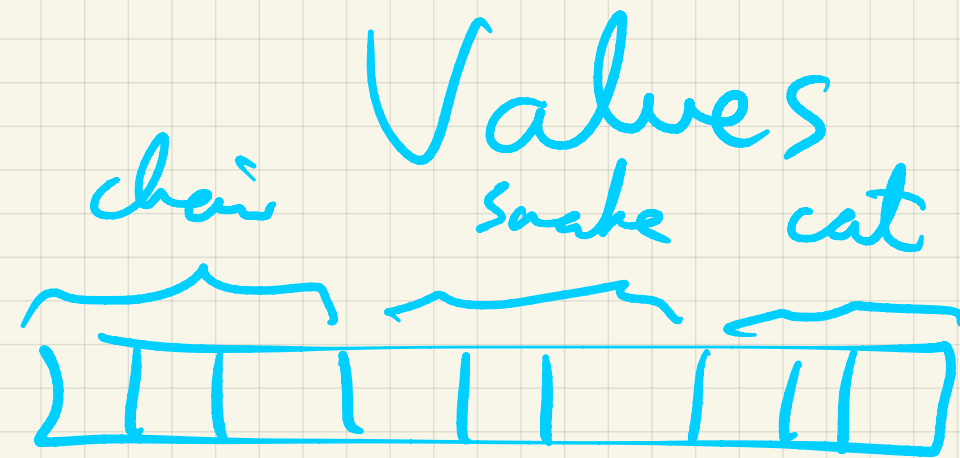
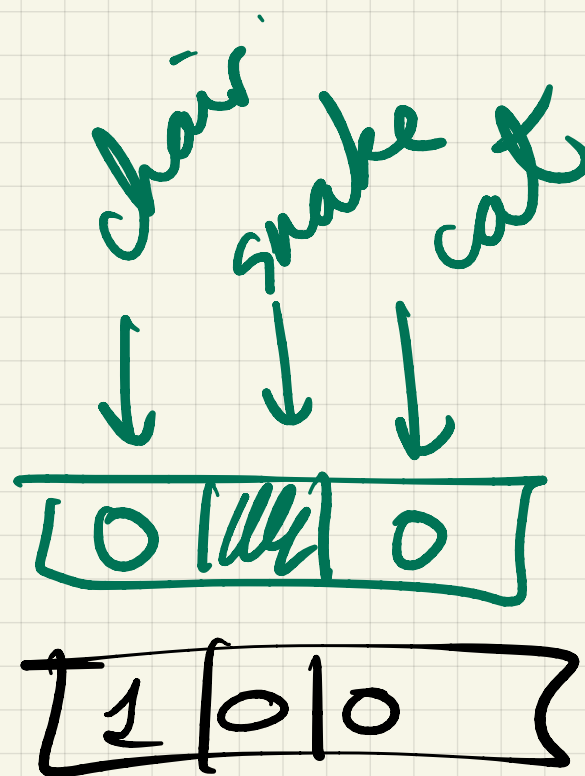
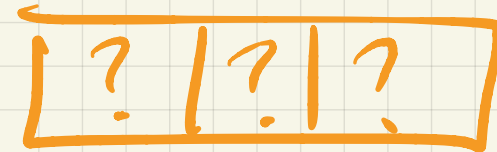
For every word



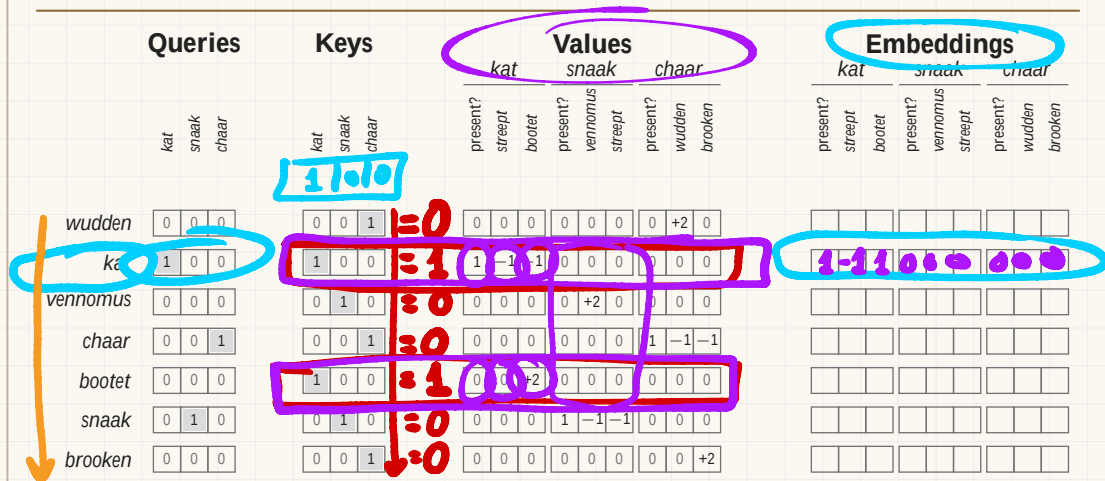
Venomous
wooden



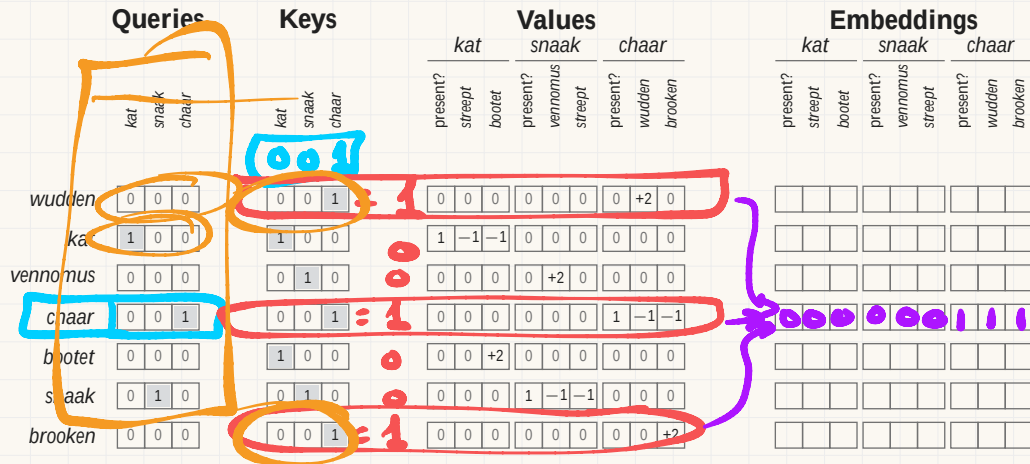
Keys



An attention map



An attention map



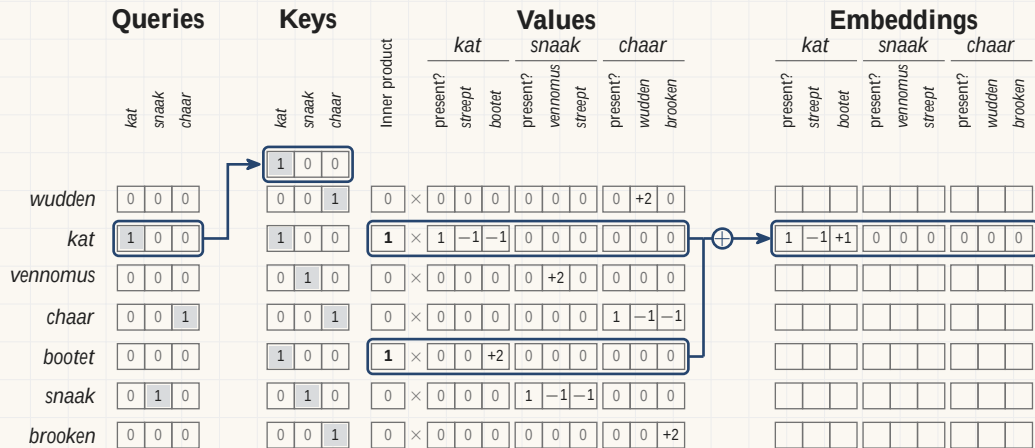
[illegible]

	Queries	Keys		Values				Embeddings																												
	<i>kat</i> <i>snaak</i> <i>chaar</i>	<i>kat</i> <i>snaak</i> <i>chaar</i>	Inner product	<i>kat</i> present? streept bootet	<i>snaak</i> present? vennomus streept	<i>chaar</i> present? wudden brooken		<i>kat</i> present? streept bootet	<i>snaak</i> present? vennomus streept	<i>chaar</i> present? wudden brooken																										
<i>wudden</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td></tr></table>	1	0	0	0	0	1	0 ×	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>			
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0	+2	0																																		
<i>kat</i>	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	1 ×	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>						
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1	0	0																																		
1	-1	-1																																		
0	0	0																																		
0	0	0																																		
<i>vennomus</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	0 ×	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>+2</td><td>0</td></tr></table>	0	+2	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>						
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0	1	0																																		
0	0	0																																		
0	+2	0																																		
0	0	0																																		
<i>chaar</i>	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	<table><tr><td>0</td><td>0</td><td>1</td></tr></table>	0	0	1	0 ×	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>						
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1	-1	-1																																		
<i>bootet</i>	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>0</td><td>0</td></tr></table>	1	0	0	1 ×	<table><tr><td>0</td><td>0</td><td>+2</td></tr></table>	0	0	+2	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>						
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0	0	+2																																		
0	0	0																																		
0	0	0																																		
<i>snaak</i>	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	<table><tr><td>0</td><td>1</td><td>0</td></tr></table>	0	1	0	0 ×	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td>1</td><td>-1</td><td>-1</td></tr></table>	1	-1	-1	<table><tr><td>0</td><td>0</td><td>0</td></tr></table>	0	0	0	<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>				<table><tr><td></td><td></td><td></td></tr></table>						
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0	1	0																																		
0	0	0																																		
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Queries			Keys			Values			Embeddings		
	kat	snaak	chaar								
	kat	snaak	chaar	kat	snaak	chaar	Inner product	kat	snaak	chaar	
								present?	streept	bootet	
wudden	0	0	0	1	0	0	$0 \times 0 \ 0 \ 0$	0	0	0	
kat	1	0	0	1	0	0	$1 \times 1 \ -1 \ -1$	0	0	0	
vennomus	0	0	0	0	1	0	$0 \times 0 \ 0 \ 0$	0	+2	0	
chaar	0	0	1	0	0	1	$0 \times 0 \ 0 \ 0$	0	0	0	
bootet	0	0	0	1	0	0	$1 \times 0 \ 0 \ +2$	0	0	0	
snaak	0	1	0	0	1	0	$0 \times 0 \ 0 \ 0$	1	-1	-1	
brooken	0	0	0	0	0	1	$0 \times 0 \ 0 \ 0$	0	0	+2	

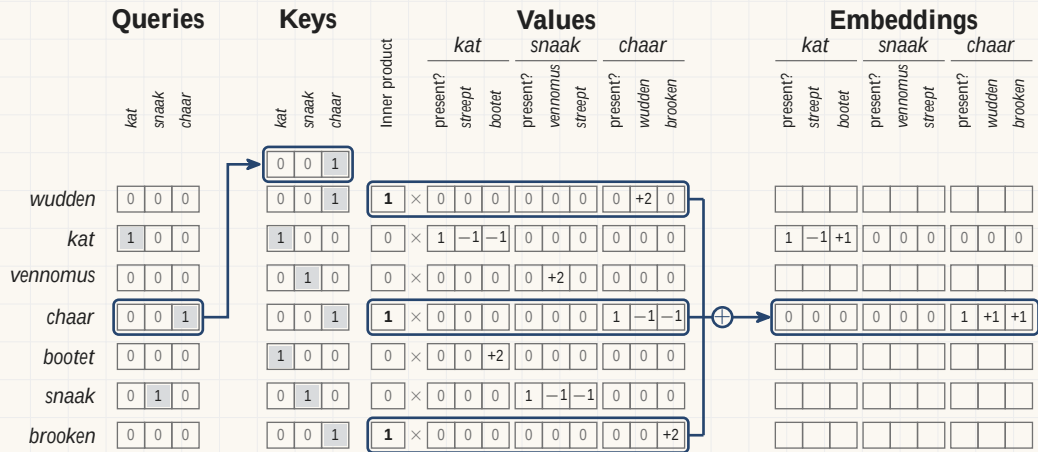
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	Queries			Keys			Inner product	Values									Embeddings														
	kat	snaak	chaar	kat	snaak	chaar		kat	snaak	chaar	present?	streept	bootet	present?	vennomus	streept	present?	wudden	brooken	kat	snaak	chaar	present?	streept	bootet	present?	vennomus	streept	present?	wudden	brooken
wudden	0	0	0	0	0	1	1	×	0	0	0	0	0	0	0	0	+2	0													
kat	1	0	0	1	0	0	0	×	1	-1	-1	0	0	0	0	0	0	0	0	1	-1	+1	0	0	0	0	0	0	0	0	0
vennomus	0	0	0	0	1	0	0	×	0	0	0	0	+2	0	0	0	0	0	0												
chaar	0	0	1	0	0	1	1	×	0	0	0	0	0	0	0	1	-1	-1													
bootet	0	0	0	1	0	0	0	×	0	0	+2	0	0	0	0	0	0	0													
snaak	0	1	0	0	1	0	0	×	0	0	0	1	-1	-1	0	0	0	0													
brooken	0	0	0	0	0	1	1	×	0	0	0	0	0	0	0	0	0	+2													

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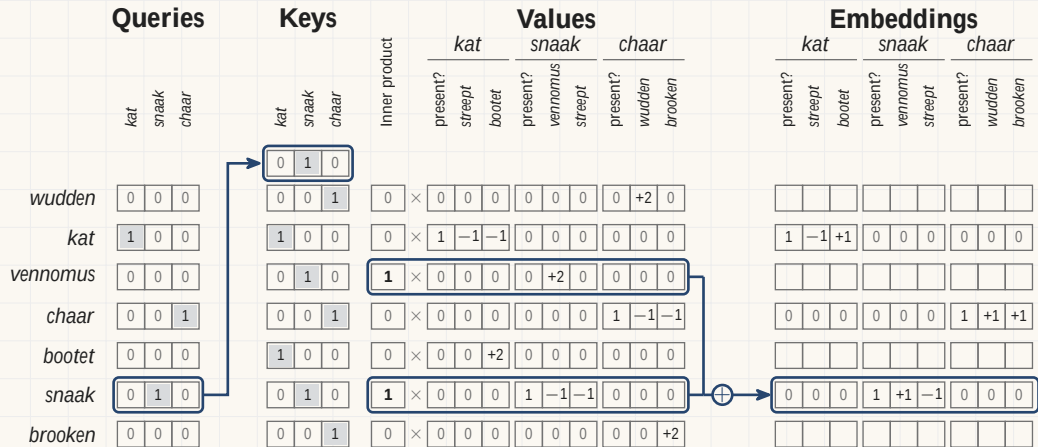


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									present?	vennomus	streept
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wudden	0	0	0		0	0	1	0 ×	0	0	0
kat	1	0	0		1	0	0	0 ×	1	-1	-1
vennomus	0	0	0		0	1	0	1 ×	0	0	0
chaar	0	0	1		0	0	1	0 ×	0	0	0
bootet	0	0	0		1	0	0	0 ×	0	0	+2
snaak	0	1	0		0	1	0	1 ×	0	0	0
brooken	0	0	0		0	0	1	0 ×	0	0	0

	kat	snaak	chaar
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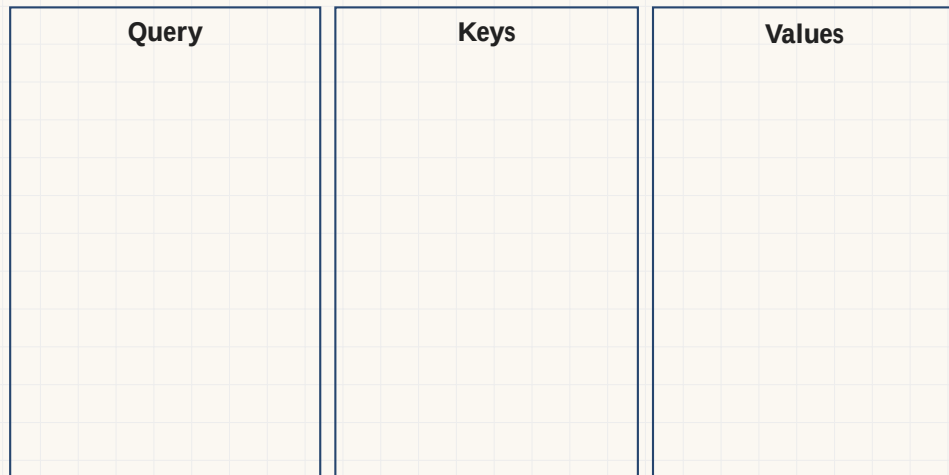
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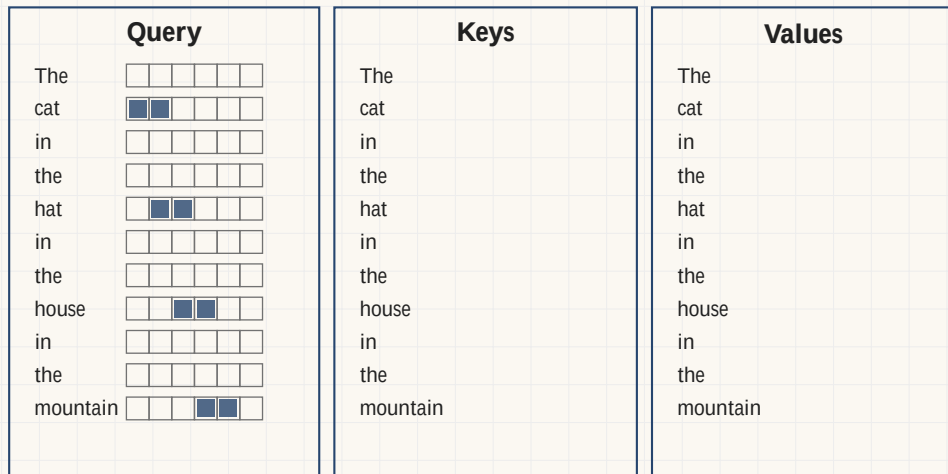
An attention map for location of objects



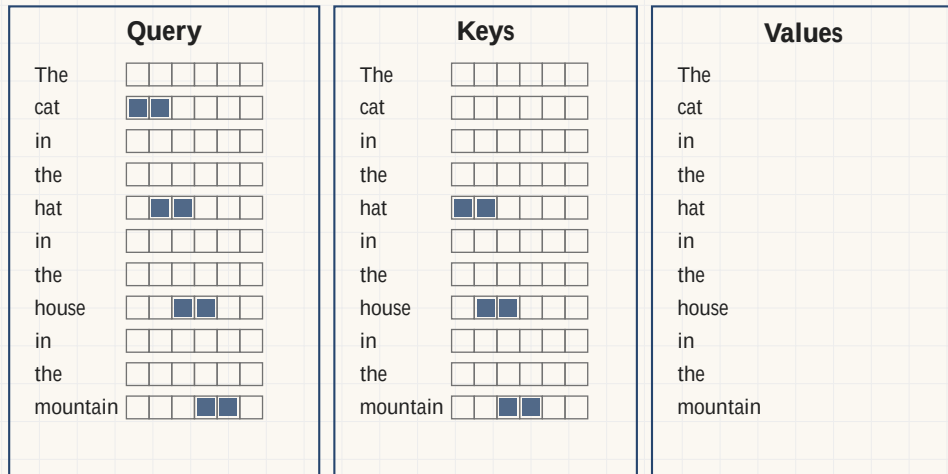
An attention map for location of objects

Query	Keys	Values
The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain	The cat in the hat in the house in the mountain

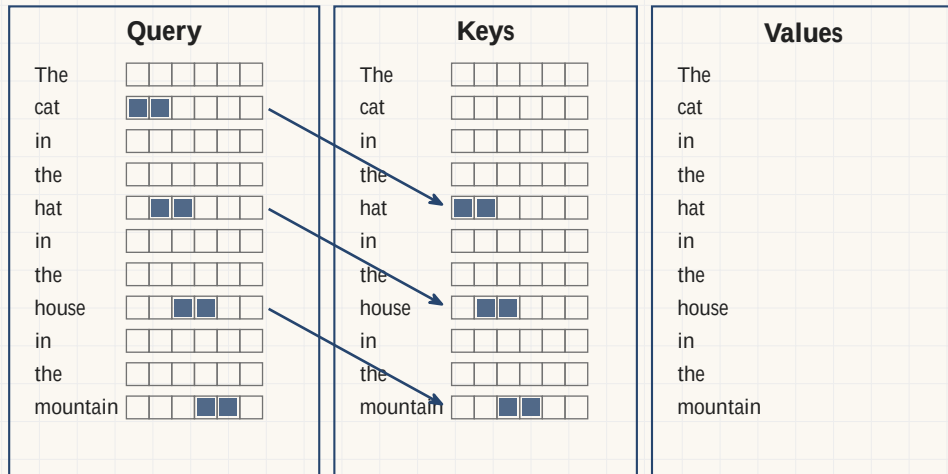
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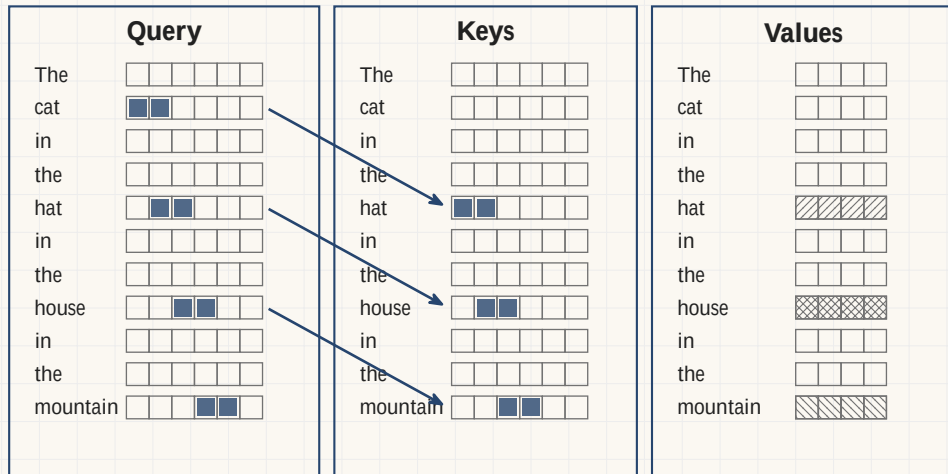
An attention map for location of objects

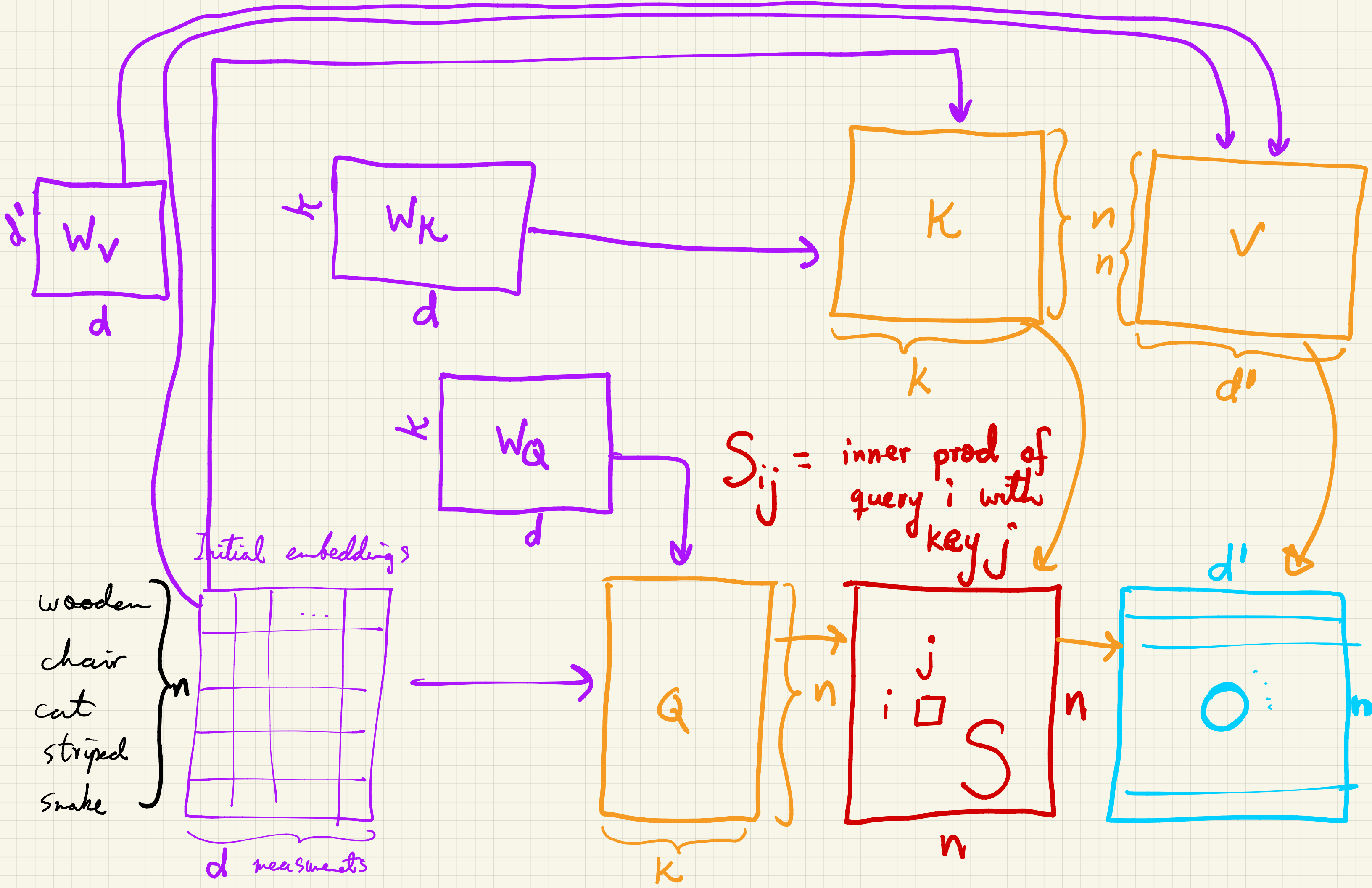


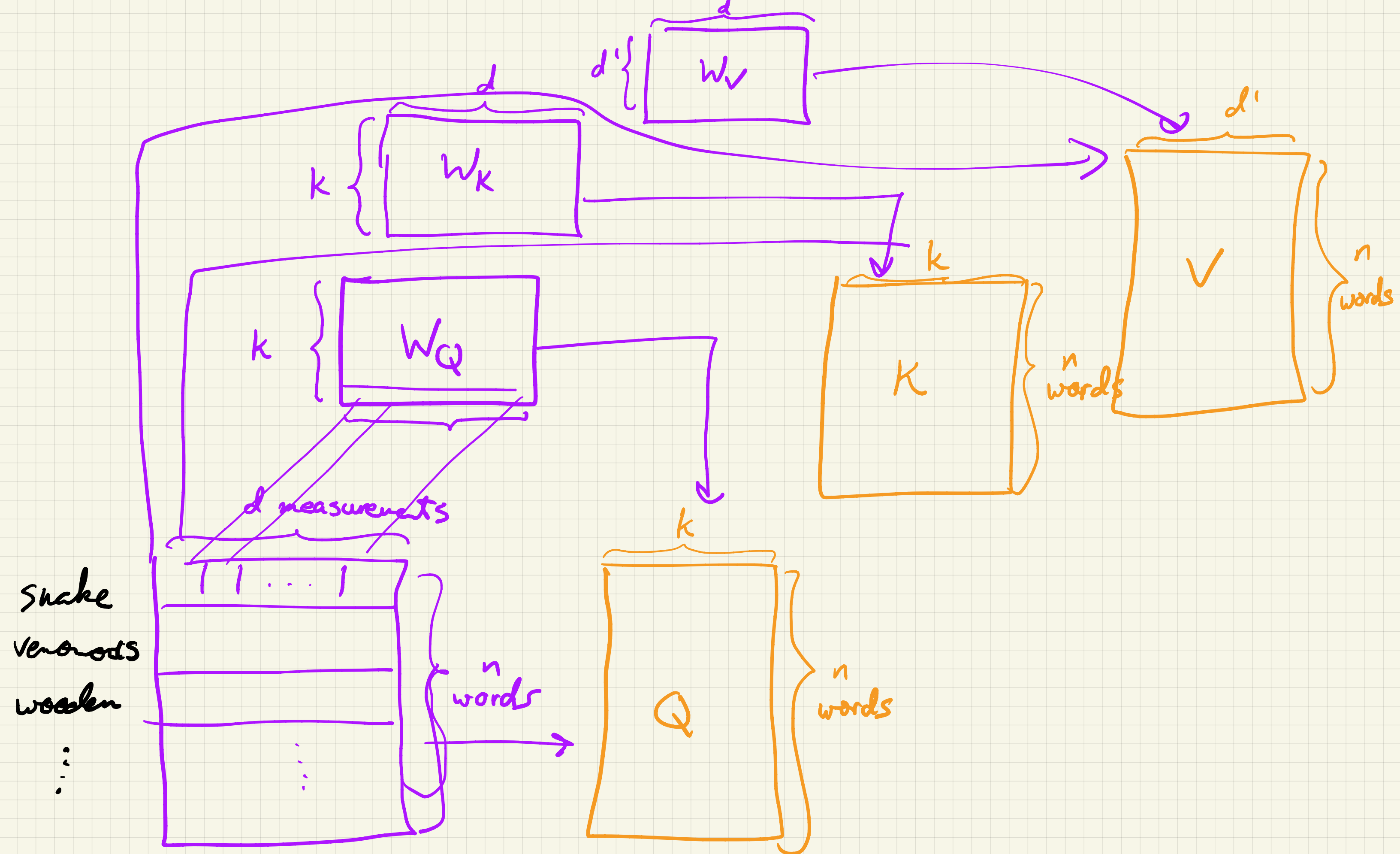
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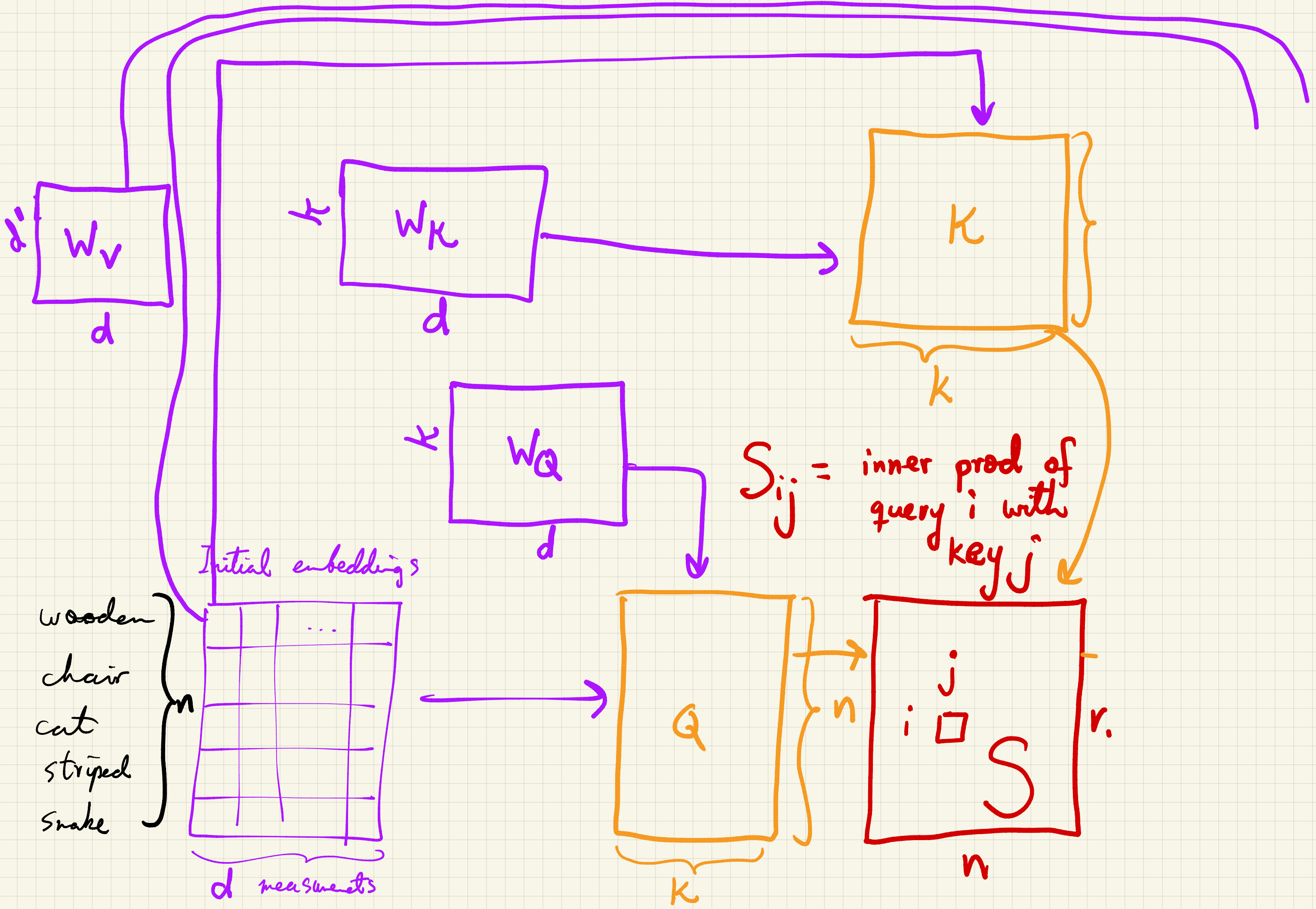


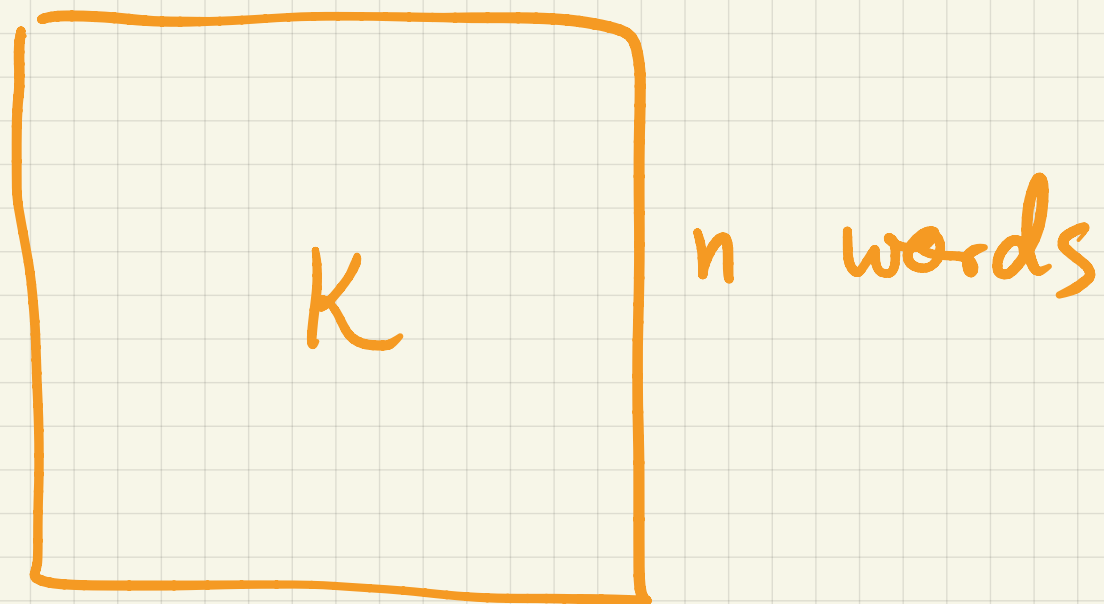
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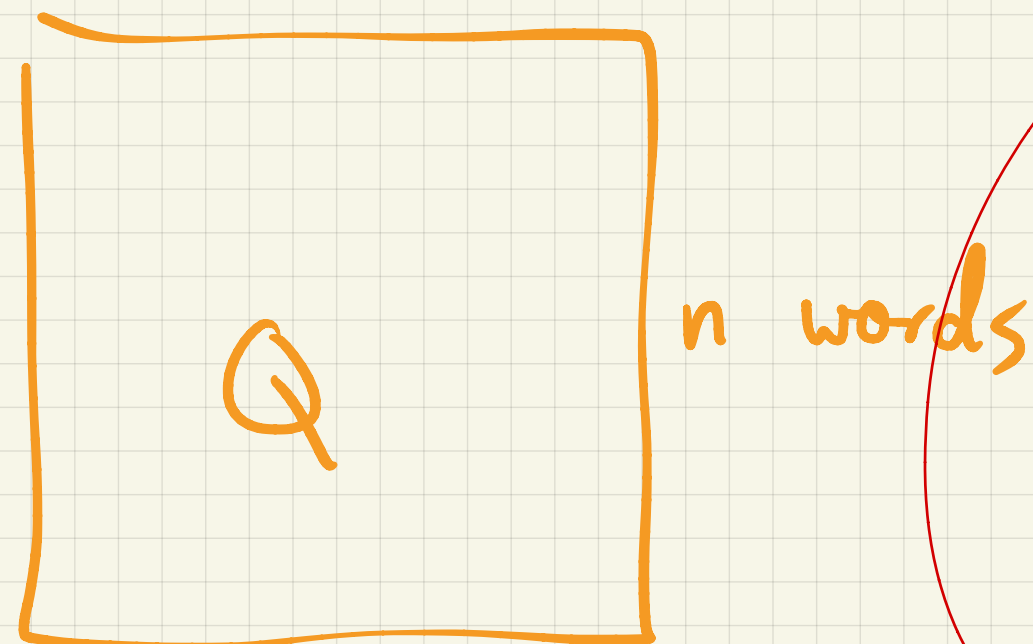




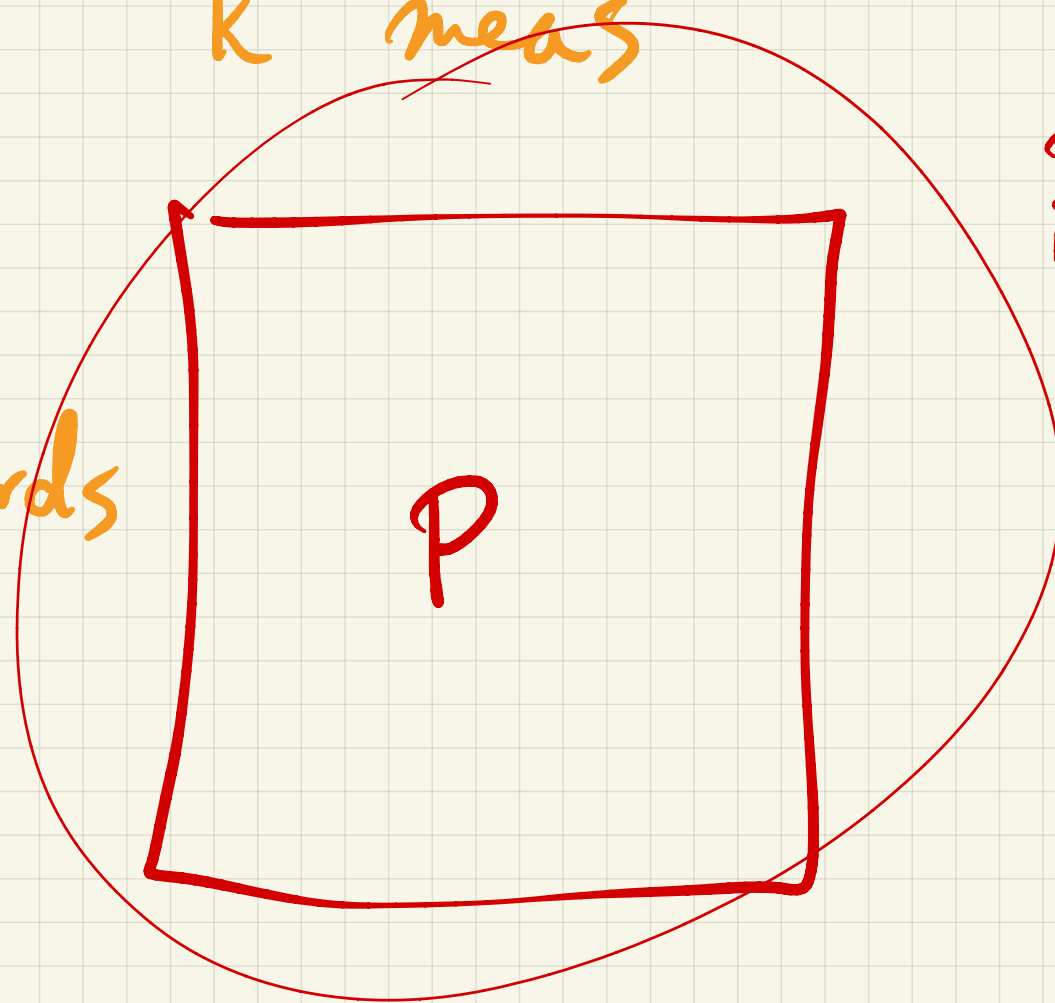




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$P_{ij} =$ inner prod
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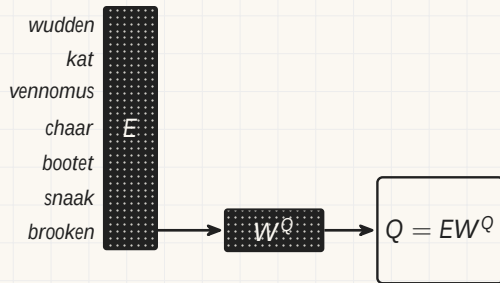
$$P = Q @ K.T$$

Abstraction of an attention mechanism

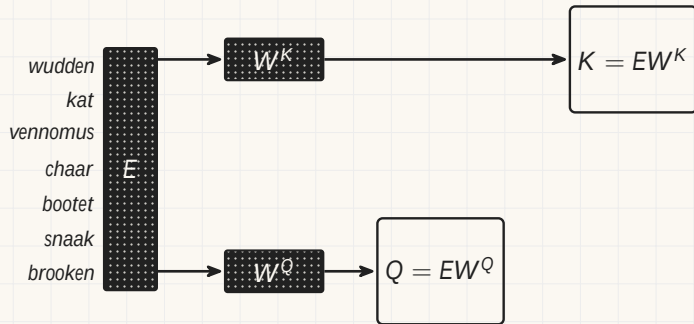
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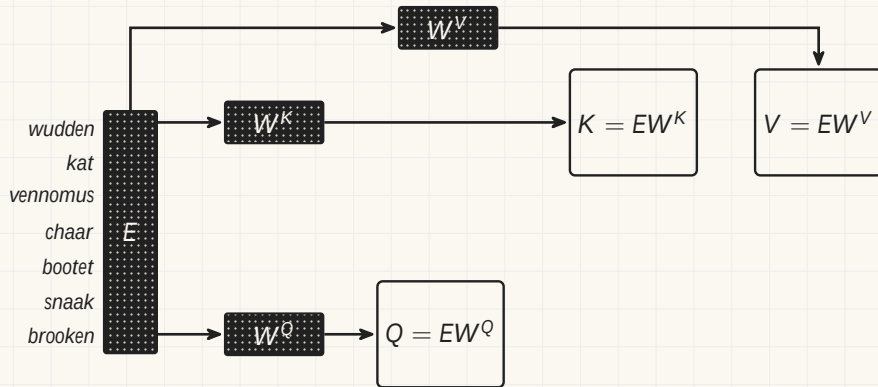
Abstraction of an attention mechanism



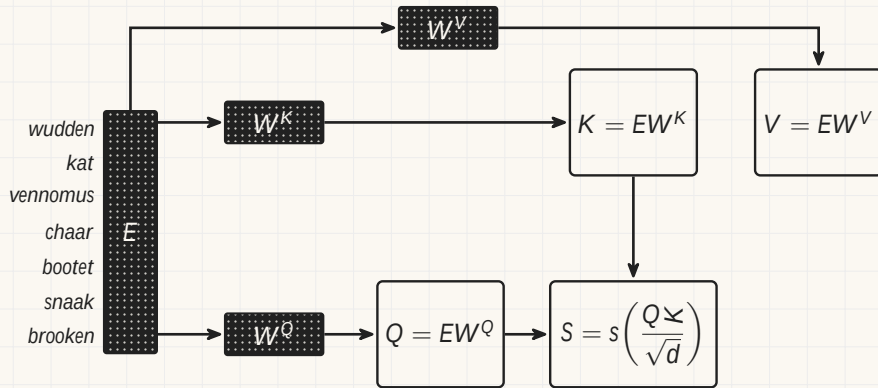
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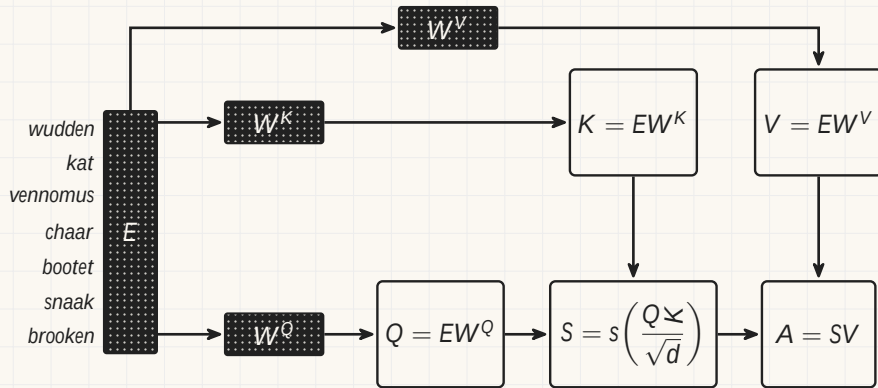
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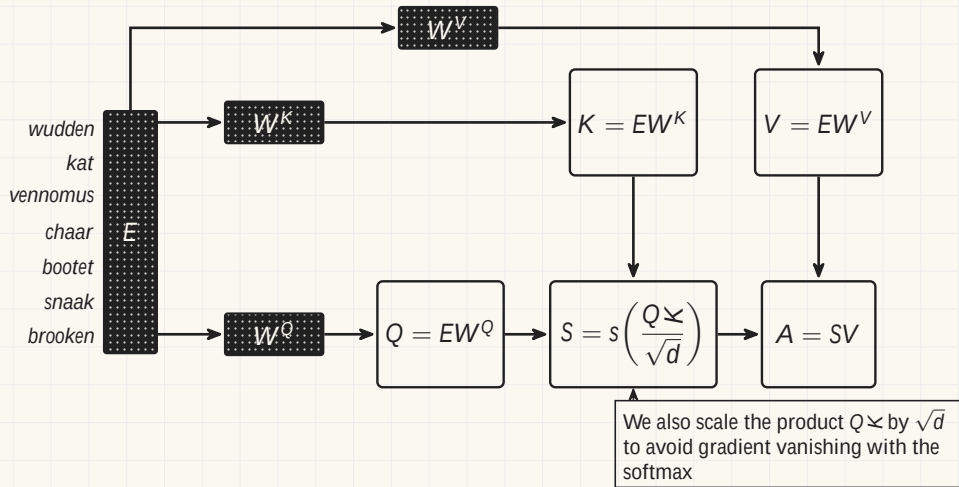
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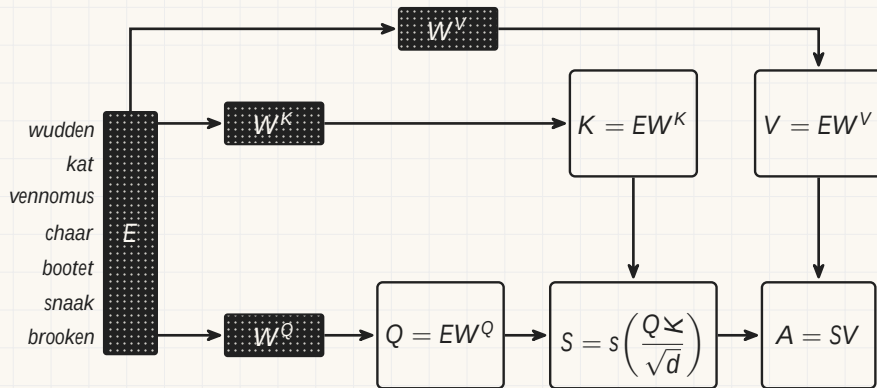
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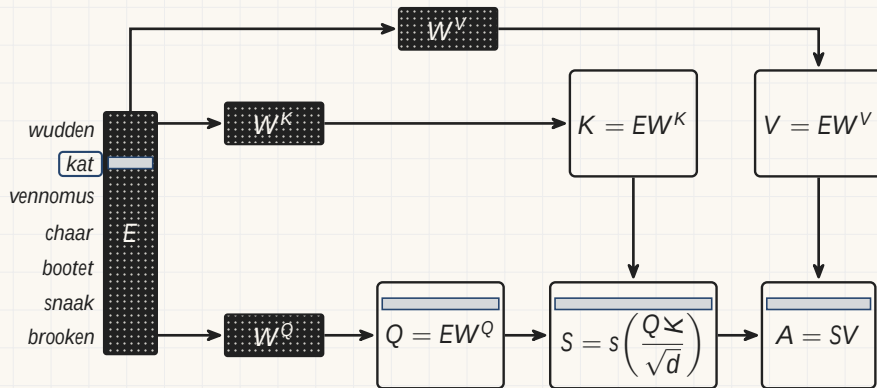
Abstraction of an attention mechanism



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval



Attention from the perspective of information retrieval

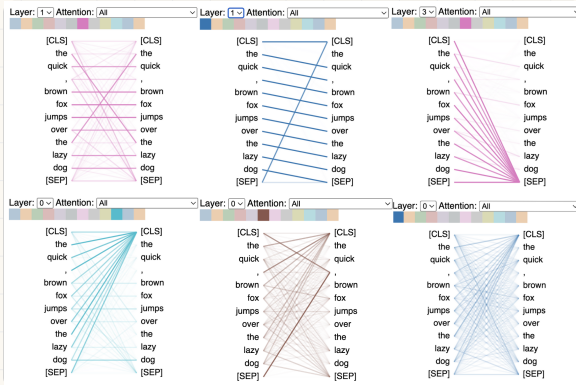


Image from: <https://www.comet.com/site/blog/explainable-ai-for-transformers/>